

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

CourseCode: 23CM4121

ExperimentNo: 1&2

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1. TITLE

Develop a Static Webpage Using HTML5 Semantic Elements, Forms and CSS3 Styling

2. AIM

To develop a static webpage using HTML5 semantic elements and CSS3, and to understand how HTML provides webpage structure while CSS provides styling and visual presentation.

3. OBJECTIVES

- To understand the basic structure of an HTML5 document.
- To use semantic HTML5 elements such as header, nav, section, main, article, aside and footer.
- To create navigation links and a user inquiry form.
- To use lists, headings, paragraphs and form controls.
- To apply CSS3 properties for layout, colors, spacing, borders and effects.
- To implement hover effects and card-based webpage design.
- To test the webpage using multiple test cases and verify the expected output.

4. THEORY

HTML5 (HyperText Markup Language) is used to create the structure and content of a webpage. Semantic elements describe the purpose of different parts of a webpage and improve readability, accessibility and maintainability.

CSS3 (Cascading Style Sheets) is used to control the presentation of HTML elements. It can be used for colors, fonts, spacing, borders, layouts, shadows, gradients and interactive hover effects.

In this experiment, the HTML program creates a semantic static webpage containing a header, navigation bar, articles, an inquiry form, an aside section and a footer. The CSS program creates a styled webpage with a gradient header, navigation styling, flexible cards, borders, shadows and hover effects.

Basic flow: HTML Structure → CSS Styling → Browser Rendering → User Interaction

5. ALGORITHM / PROCEDURE

- Create a new HTML file and write the HTML5 document structure.
- Add a header containing the website title and navigation links.
- Create the home section using section and paragraph elements.
- Create the main content area using main and article elements.
- Add a user inquiry form using form, fieldset, legend, label, input, select and textarea elements.
- Create an aside section containing benefits and components using unordered and ordered lists.
- Add a footer to complete the webpage.

- Create/apply CSS3 styles for the header, navigation, cards, buttons and footer.
- Add hover effects for navigation links, cards and buttons.
- Open the webpage in a browser and verify its appearance and functionality.
- Perform multiple test cases and compare expected and actual results.

6. PROGRAM – HTML5

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Static Webpage</title>
5
6   <!-- CSS File -->
7   <link rel="stylesheet" href="style.css">
8 </head>
9
10 <body>
11
12   <!-- Header -->
13   <header>
14     <h1>StaticWeb</h1>
15
16     <!-- Navigation -->
17     <nav>
18       <a href="#home">Home</a>
19       <a href="#articles">Articles</a>
20       <a href="#about">About</a>
21       <a href="#contact">Contact</a>
22     </nav>
23   </header>
24
25   <!-- Home -->
26   <section id="home">
27     <h2>Semantic HTML Webpage</h2>
28
29     <p>
30       This webpage is created using HTML5 and CSS3.
31     </p>
32
33     <a href="#articles">Read Articles</a>
34   </section>
35
36   <!-- Main Content -->
37   <main id="articles">
```

.vscode > > index.html > html > body > header > nav

```
2  <html>
10 <body>

36 <!-- Main Content -->
37 <main id="articles">
38
39   <article>
40     <h2>1. Core Structural Tags</h2>
41
42     <p>Published: July 7, 2026</p>
43
44     <p>
45       HTML provides semantic tags like
46       main, article and section to organize
47       webpage content.
48     </p>
49
50     <p>
51       These tags make the webpage easy to
52       understand and accessible.
53     </p>
54   </article>
55
56   <article>
57     <h2>2. User Inquiry Form</h2>
58
59     <p>Published: July 5, 2026</p>
60
61     <form id="contact">
62       <fieldset>
63         <legend>User Inquiry Form</legend>
64
65         <label>Name:</label>
66         <input type="text" required>
67         <br><br>
68       </fieldset>
69     </form>
  
```

✓ Widely available across major browsers (Baseline since 2015)

< MDN Reference

<input type="email" required>

<label>Topic:</label>

<select>

<option>General Feedback</option>

<option>Technical</option>

<option>Academics</option>

</select>

<label>Message:</label>

<textarea rows="5" cols="30"></textarea>

<input type="submit" value="Submit">

<input type="reset" value="Clear">

</fieldset>

</form>

</article>

</main>

```
102     <aside id="about">
103         <h3>About</h3>
104
105         <p>
106             Static webpages display information
107             without a database.
108         </p>
109
110         <h4>Benefits</h4>
111
112         <ul>
113             <li>Fast loading</li>
114             <li>Low memory usage</li>
115             <li>Easy to understand</li>
116         </ul>
117
118         <h4>Components</h4>
119
120         <ol>
121             <li>HTML Document</li>
122             <li>Semantic Tags</li>
123             <li>Form Elements</li>
124         </ol>
125     </aside>
126
127     <!-- Footer -->
128     <footer>
129         <p>
130             © 2026 StaticWeb. Created using HTML and CSS.
131         </p>
132     </footer>
133
134 </body>
135 </html>
```

6. PROGRAM – CSS3

```
1  body {
2      margin: 0;
3      font-family: Arial;
4      background: #f2f2f2;
5  }
6
7  header {
8      padding: 25px;
9      text-align: center;
10     color: white;
11     background: linear-gradient(to right, royalblue, purple);
12 }
13
14 nav {
15     padding: 15px;
16     text-align: center;
17     background: #222;
18 }
19
20 nav a {
21     color: white;
22     margin: 15px;
23     text-decoration: none;
24 }
25
26 nav a:hover {
27     color: yellow;
28 }
29
30 section {
31     text-align: center;
32     padding: 30px;
33 }
34
35 main {
36     padding: 20px;
37 }
```

```
38
39 article {
40     background: ■ white;
41     margin: 20px;
42     padding: 20px;
43     border: 2px solid ■ royalblue;
44     border-radius: 10px;
45     box-shadow: 3px 3px 8px ■ gray;
46 }
47
48 aside {
49     background: ■ white;
50     margin: 20px;
51     padding: 20px;
52     border-radius: 10px;
53 }
54
55 input,
56 select,
57 textarea {
58     padding: 8px;
59 }
60
61 input[type="submit"],
62 input[type="reset"] {
63     padding: 8px 15px;
64     background: ■ royalblue;
65     color: ■ white;
66     border: none;
67     border-radius: 5px;
68 }
69
70 input[type="submit"]:hover,
71 input[type="reset"]:hover {
72     background: □ black;
73 }
```

```
60
61 input[type="submit"],
62 input[type="reset"] {
63     padding: 8px 15px;
64     background: #000080royalblue;
65     color: #FFFFFFwhite;
66     border: none;
67     border-radius: 5px;
68 }
69
70 input[type="submit"]:hover,
71 input[type="reset"]:hover {
72     background: #000000black;
73 }
74
75 footer {
76     padding: 15px;
77     text-align: center;
78     background: #222222;
79     color: #FFFFFFwhite;
80 }
```


7. CODE EXPLANATION

Code / Element	Explanation
<header>	Defines the header area of the webpage.
<nav>	Contains navigation links.
<section>	Groups related webpage content.
<main>	Defines the main content of the page.
<article>	Represents independent content such as an article.
<form>	Creates the user inquiry form.
<fieldset> and <legend>	Groups and labels related form controls.
<aside>	Contains additional information such as benefits and components.
<footer>	Defines the footer area of the webpage.
display: flex	Creates a flexible layout for the cards.
background / gradient	Applies background colors and gradient styling.
border-radius	Creates rounded corners.
box-shadow	Adds a shadow around cards.
:hover	Applies styling when the pointer is placed over an element.
transform: scale()	Slightly enlarges the card during hover.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Home navigation link	Home link is present and clickable	Home link displayed correctly	Pass
TC-02	Semantic HTML structure	Header, nav, section, main, article, aside and footer are displayed	Semantic sections displayed correctly	Pass
TC-03	User inquiry form	Name, Email, Topic and Message fields are available	Form fields displayed correctly	Pass
TC-04	Required field validation	Submit form with Name and Email empty	Browser requests required fields	Pass
TC-05	CSS navigation hover	Move mouse over navigation link	Link color changes on hover	Pass
TC-06	CSS card hover	Move mouse over a card	Card slightly enlarges	Pass
TC-07	Form reset	Enter data and click Clear	Entered form data is cleared	Pass

9. OUTPUT

- Output 1: HTML static webpage with header, navigation, semantic content, inquiry form, aside and footer.
- Output 2: CSS3 styled webpage with gradient header, navigation bar and three cards.
- Output 3: Hover effects are visible on navigation links, cards and buttons.
- Output 4: Required form validation works when mandatory fields are left empty.
- Output screenshots should be inserted here after actual browser execution.
-

10. RESULT

Thus, a static webpage was successfully developed using HTML5 semantic elements and CSS3. The webpage structure, navigation, inquiry form, lists, styling, responsive viewport setting and hover effects were implemented and verified using multiple test cases.

Semantic HTML Webpage

This webpage is created using HTML5 and CSS3.

[Read Articles](#)

1. Core Structural Tags

Published: July 7, 2026

HTML provides semantic tags like main, article and section to organize webpage content.

These tags make the webpage easy to understand and accessible.

2. User Inquiry Form

Published: July 5, 2026

2. User Inquiry Form

Published: July 5, 2026

User Inquiry Form

Name:

Email:

Topic:

Message:

About

Static webpages display information without a database.

Benefits

- Fast loading
- Low memory usage
- Easy to understand